Project Proposal

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Abstract

The project aims to develop a goal assignment system for robots through voice commands, converting natural language to machine language by using propositional logic which will be integrated into automated planning algorithms.

This proposal is driven by the challenge of designing an interface between humans and robots using a set of artificial intelligence algorithms, including Large Language Models, Speech-To-Text, and Automated Planning models.

The project will be divided into different phases. Research and requirements definition, global design and system development, test suite design and final validation tests. It is expected to deliver a functional prototype working in a test environment.

KeyWords

Natural Language Processing, Automated Planning, Large Language Model, Artificial Intelligence

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1 Project Description

1.1 Introduction

Human-Machine interaction aims to establish communication tools between humans and machines. The goal of this technology is to create a simple and natural way of communication, seeking to simplify robot handling. Traditional robots were restricted to repetitive, predefined tasks within controlled environments. By integrating Artificial Intelligence, robots can manage a broader range of tasks that were once considered too complex, enabling them to make intelligent, real-time decisions. To achieve this, my proposal combines Symbolic and Sub-Symbolic AI. Sub-Symbolic AI will enable the system to process and interpret Natural Language inputs from users and, while Symbolic will employ logic-based reasoning to generate action plans.

1.2 Objective

The aim of this project is to design a system that interprets voice commands in English and converts them into goals represented using propositional logic. These goals will be used to solve tasks through automated planning. The project consists of the following elements:

- Speech Recognition System: The system will receive and interpret voice commands in English using a Speech-To-Text algorithm.
- Natural Language Processing: The system will extract the initial state, the final state and the environmental constraints from the message, using a Large Language Model convert it into PDDL.
- Automated Planning Algorithm: The system will generate a set of actions to reach a goal state from an initial state, using Automated Planning.
- Web Environment: The project will initially be deployed in a web environment for prototyping.

The expected outcome is to deliver a tool capable of receiving, interpreting, and solving a problem in a test environment.

1.3 Technical Requirements

To develop the project, I will need to:

- Identify and integrate a suitable Speech-to-Text model to implement it into the system. The approach involves using a third-party API to send the audio file and receive the transcribed message.
- Create a Client HMI that allows the user to record and send the audio file to be interpreted. Initially, the interface will be built using Streamlit as a demo, but the final version will be implemented using Flutter.
- Prompt Fine-Tune an existing LLM model like Llama 3.1 or Mistral to extract the information following a pre-established structure.
- Automated Planning Algorithm to build the set of actions that the robot needs to perform. This will be implemented using Fast Downward, an automated planning system that generates plans.

1.4 Use Case

This section presents a use case demonstrating how the system will operate. In this scenario, the user acts as the main actor recording a voice message. This message is sent to a Speech-to-Text service, which generates the transcribed text. Once the system has the natural language message, it will be processed by a LLM to obtain the initial and goal states in PDDL format. Finally, this problem, along with the predefined domain, will be sent to the planner to define the action set for use in a simulated environment. As shown in Figure 1.

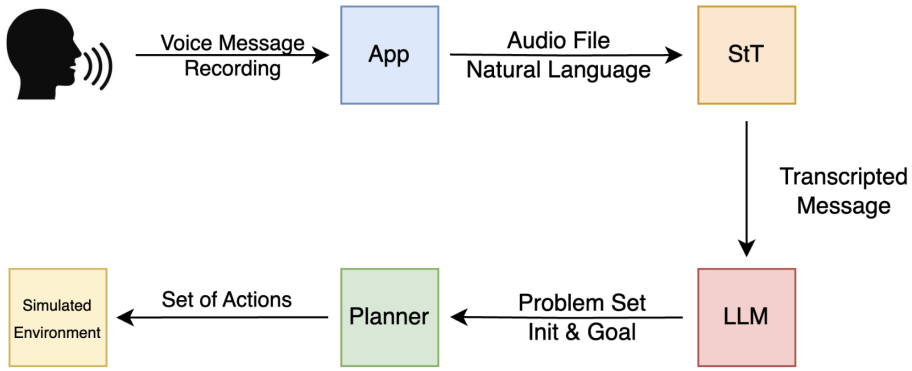


Figure 1: Use Case Diagram.

1.5 System Architecture

The system will be built following microservices architecture, see Figure 2. Server will be responsible for handling and coordinating the communication with the LLM, the Planner and the Speech-to-Text service.

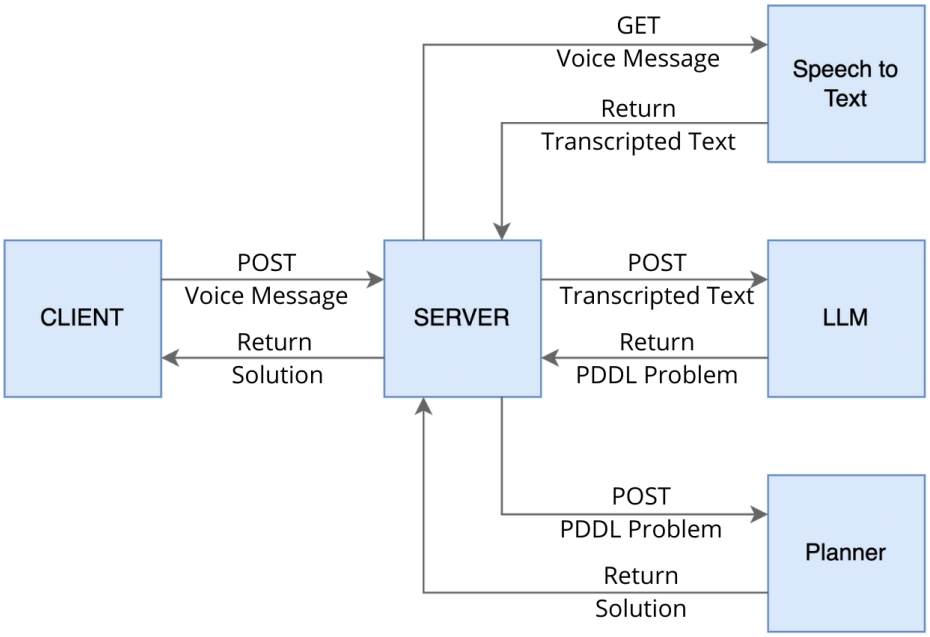


Figure 2: WorkFlow Diagram. System Architecture.

2 Motivation

This project aligns with my interest in Artificial Intelligence, and I am keen to acquire these skills as they will be essential for my professional career. For this reason, I have chosen a project that involves multiple AI models, covering different development phases: from the implementation of pre-trained models, retraining already existing models to the creation of new models from scratch.

Additionally, being part of a research project fosters a strong sense of belonging to a team that is working on something larger. This emphasizes the need to produce high-quality work that meets the standards of the team and the research itself.

Lastly, developing this project with my tutor, Moisés Martínez, allows me not only to learn from his expertise but also to take on the challenge of meeting his high expectations, pushing myself to improve and deliver a high-quality project.

3 Expected Results and Feasibility

This section includes the expected results of the project, the development timeline, the current knowledge, the required knowledge to be acquired, and the courses involved.

3.1 Expected Results

A simulated environment is expected to be delivered, allowing the user to record a voice message, which will be converted to text. An LLM model will be capable of extracting relevant information from the message: the initial state, the final state, and the system constraints. Once this information is extracted, the system will automatically generate the necessary PDDL files to be processed by an Automated Planning Algorithm. The Human-Machine interface will be displayed in a web environment that allows the user to follow the process.

3.2 Estimated Time For Technical Development

The development stages of this project are divided into five components, each with its respective tasks. Time allocation for each stage is outlined in Table 1.

Component	**Tasks**	**Initial Date**	**Initial Date**
Project Definition	Previous Research	02/09/2024	20/09/2024
	Project Scope Definition	23/09/2024	07/10/2024
	Project Design	08/10/2024	31/10/2024
Project Development	Speech-to-Text	01/11/2024	21/11/2024
	Prompt Fine Tuning	22/11/2024	23/12/2024
	Planner	24/12/2024	28/02/2025
	Front-End	03/03/2025	14/03/2025
	Component Integration	17/03/2025	31/03/2025
Project Deployment	Deployment	01/04/2025	07/04/2025
	Validation Tests	08/04/2025	21/04/2025
Documentation	Report	03/02/2025	30/05/2025
	Presentation	15/05/2025	30/05/2025
Tutoring Sessions		02/09/2024	30/05/2025

Table 1: Project Schedule

The following Gantt's Diagram is showing how tasks are distributed. Figure 3.

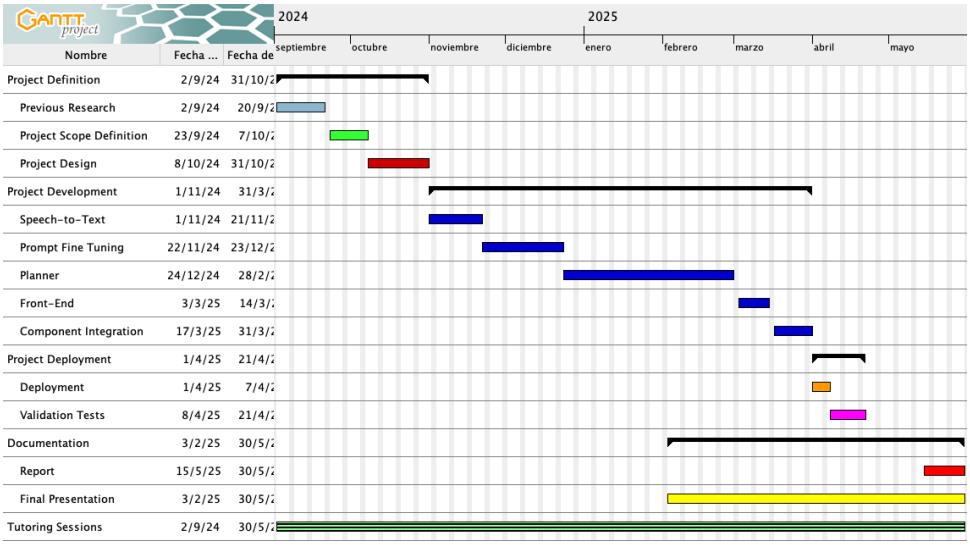


Figure 3: Gantt's Diagram [1]

4 Bibliography

References

[1] *Ganttproject: Project management tool.* 2024. [Online]. Available: `https://www.ganttproject.biz/`.